



Lithosyntrophy: Obligate syntrophy in a phosphite-oxidizing, methanogenic culture

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The anaerobic conversion of organic matter to methane and carbon dioxide typically relies on obligate syntrophic interactions between bacteria and methanogenic archaea, where interspecies electron transfer enables thermodynamically constrained reactions to proceed near equilibrium. Syntrophs often couple the oxidation of fermentation products such as fatty acids and alcohols to the reduction of protons to form hydrogen (H_2). These reactions can only proceed if low H_2 concentrations are maintained by H_2 -consuming syntrophic partners. Here, we describe “lithosyntrophy,” a mode of syntrophic interaction in which electrons that drive hydrogenotrophic methanogenesis originate from an inorganic compound rather than from the canonical organic substrates. *Candidatus* Phosphitivorax anaerolimi Phox-21 oxidizes phosphite (HPO_3^{2-} , oxidation state +3) to phosphate coupled to hydrogenogenesis in an obligate energetic dependency on a hydrogenotrophic methanogen, *Methanoculleus* sp. Physiology experiments, thermodynamic calculations, genomic annotation, and metaproteomics analysis collectively revealed a mechanism for syntrophic phosphite oxidation in Phox-21. In this pathway, electrons derived from phosphite drive H_2 production via an electron-confurcating hydrogenase. Unlike previously characterized acetogenic phosphite oxidizers, Phox-21 is a mixotroph that assimilates acetate to form biomass. Lithosyntrophic phosphite oxidizers may play important roles both in transferring reducing equivalents as well as biologically available phosphorus to other members of their surrounding microbial communities, establishing a previously unrecognized metabolic and biogeochemical link between the phosphorus and carbon redox cycles in anoxic ecosystems.

dissimilatory phosphite oxidation | anaerobes | syntrophy | lithosyntrophy | methanogenesis

In anoxic environments, the degradation of organic matter to methane and carbon dioxide involves obligate syntrophic interactions between bacteria and methanogenic archaea (1–3). Obligate syntrophy is defined by a metabolic and thermodynamic interdependence, where the complete degradation of a substrate requires the activity of two or more microorganisms. Neither syntrophic partner can degrade the organic compound alone, and both catalyze catabolic reactions that proceed close to thermodynamic equilibrium, such that the energetics of one partner’s catabolism is directly influenced by the other. In other words, the accumulation of end products from one organism’s metabolism can thermodynamically inhibit the reaction, and the syntrophic partner drives the reaction forward by removing inhibitory end products. In organotrophic syntrophic partnerships (“organosyntrophy”), syntrophic bacteria use diverse forms of interspecies electron transfer in the form of interspecies formate transfer, direct interspecies electron transfer (DIET), quinone-mediated interspecies electron transfer (QUIET), and interspecies hydrogen (H_2) transfer to overcome these thermodynamic constraints (3, 4). Organosyntrophs involved in interspecies H_2 transfer couple the oxidation of organic acids and alcohols to the reduction of protons to form H_2 , which is immediately consumed by a hydrogenotrophic methanogen (5–8). The drawdown of H_2 by the methanogen enables the otherwise endergonic (thermodynamically unfavorable, $\Delta G_r > 0$) catabolic reaction of the syntrophic bacterium to become exergonic (thermodynamically favorable, $\Delta G_r < 0$).

While obligate syntrophy has traditionally been associated with the anaerobic degradation of organic substrates (1, 2, 5, 7), the underlying thermodynamic principles are not limited to organotrophy. In principle, any oxidation reaction that produces an inhibitory accumulation of H_2 may require syntrophic coupling to proceed. However, such interactions involving inorganic electron donors have not been previously described. While H_2 formation from FeS and H_2S (Wächterhäuser reaction) has been shown to support hydrogenotrophic methanogenesis, the biological mechanism supporting this reaction is unclear (9, 10), and

Significance

Dissimilatory phosphite-oxidizing microorganisms (DPOM) use phosphite as an energy source, producing phosphate. While the two previously isolated DPOM couple phosphite oxidation to carbon fixation via the Wood–Ljungdahl pathway, *Candidatus* Phosphitivorax anaerolimi Phox-21 instead performs lithosyntrophic metabolism, coupling DPO to proton reduction in obligate partnership with a hydrogenotrophic methanogen. This establishes a link between the phosphorus and carbon redox cycles: like organosyntrophy, DPO can fuel methanogenesis. Lithosyntrophy expands our understanding of syntrophic interactions and suggests that similar processes may occur with other inorganic substrates in anoxic ecosystems.

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abiotic H₂ formation from FeS has been shown to occur (11). Here, we report the identification of phosphite-driven lithosyntrophy, a mode of obligate syntrophic interaction in which electrons that drive hydrogenotrophic methanogenesis originate from an inorganic compound rather than from an organic substrate.

This finding builds on prior studies of dissimilatory phosphite oxidation (DPO), a microbial catabolism by which phosphite (HPO₃²⁻, oxidation state +3) is oxidized to phosphate. Since the first demonstration of DPO in 2000 (12), only two DPO microorganisms (DPOM) have been isolated in pure culture. *Desulfotignum phosphitoxidans* FiPS-3 and *Phosphitispora fastidiosa* DY19 were isolated from enrichments that used phosphite as the sole electron donor, CO₂ as the sole carbon source, and sulfide as a reducing agent (13, 14). In these enrichments, CO₂ also served as the electron acceptor for DY19, while sulfate was included as the electron acceptor for FiPS-3. Both isolates couple their energy metabolism and carbon metabolism by using phosphite as an electron donor for CO₂ fixation through the Wood–Ljungdahl pathway, which is the only carbon fixation pathway that results in energy conservation (15). FiPS-3 can additionally couple phosphite oxidation to dissimilatory sulfate reduction. In FiPS-3 and DY19, acetate is produced as a detectable byproduct (13, 16).

Candidatus Phosphitivorax anaerolimi Phox-21, the third described DPOM, was enriched from wastewater treatment sludge in chemically defined media that contained phosphite as the primary electron donor, CO₂/bicarbonate as the exogenous electron acceptor and cysteine as a reducing agent (17, 18). Based on its genomic potential for the reductive glycine pathway, and its dependency on CO₂ for growth, it was initially assumed that, like FiPS-3 and DY19, Phox-21 was an autotroph coupling DPO to carbon fixation (17, 18). The reductive glycine pathway shares much of the methyl branch of the Wood–Ljungdahl pathway but lacks the carbonyl branch. Instead, the reductive glycine pathway condenses methylene-THF with ammonia and CO₂ to form glycine, which can be converted to serine and then to pyruvate (17, 18). However, several lines of evidence suggested that Phox-21 is physiologically distinct from FiPS-3 and DY19. Phox-21 was resistant to isolation via dilution-to-extinction or colony picking from agar plugs, which is consistent with its phylogenetic relatedness to fastidious syntrophic bacteria within

the phylum Desulfobacterota (17, 18). In contrast to FiPS-3, Phox-21 lacks an electron transport chain and pathways for alternative metabolic strategies such as anaerobic respiration or fermentation, which are characteristics of metabolically specialized syntrophs (1). Furthermore, Phox-21 lacks genes for phosphoacetyltransferase (PTA) and acetate kinase (ACK) within the reductive glycine pathway (17), preventing acetate production and ATP generation via substrate-level phosphorylation (19). Without energy conservation from the reductive glycine pathway, it was unclear what the terminal electron acceptor for phosphite was. Collectively, these genomic, metabolic, and phylogenetic features suggested that Phox-21 is physiologically distinct from isolated DPOM and depends on syntrophic interactions with other microorganisms to oxidize phosphite.

In this study, we describe lithosyntrophic phosphite oxidation, an uncharacterized mode of obligate syntrophy in which Phox-21 couples the oxidation of phosphite to proton reduction, generating H₂ that is consumed by a hydrogenotrophic methanogen, *Methanoculleus* sp (Table 1, Rxns. 1 and 2). By demonstrating that inorganic compounds can sustain methanogenic ecosystems, these findings establish a mechanistic link between the phosphorus and carbon redox cycles and challenge the assumption that obligate syntrophy is inherently linked to organic carbon degradation (1, 2). Similar interactions with other inorganic compounds may be more widespread in contemporary anoxic environments than previously appreciated and warrants further investigation. This work provides insights into the thermodynamic and ecological constraints that shape microbial community structure and function in methanogenic environments and raises questions about the environmental distribution and evolutionary origins of phosphorus-based energy metabolisms.

Results

Electrons from Phosphite Are Stoichiometrically Balanced with Methanogenesis or Sulfate Reduction. As of July 2024, highly enriched phosphite-oxidizing cultures maintained with cysteine as a reducing agent supported a stable microbial community comprising 16 taxa, including *Candidatus* Phosphitivorax anaerolimi Phox-21, *Methanoculleus* sp., *Acetomicrobium hydrogeniformans*, and other

Table 1. Metabolic reactions involved in the lithosyntrophic phosphite oxidizing culture, their standard state Gibbs energy yields (ΔG°) at 37 °C, and their Gibbs energy yields (ΔG) under in vitro conditions

Number	Reaction	Description	ΔG° (kJ/mol rxn)	ΔG (kJ/mol rxn)
1	$HPO_3^{2-} + H_2O \rightarrow HPO_4^{2-} + H_{2(aq)}$	Lithosyntrophic DPO	−21.79	−52.91
2	$4HPO_3^{2-} + HCO_3^- + H_2O + H^+ \rightarrow 4HPO_4^{2-} + CH_{4(aq)}$	Lithosyntrophic DPO coupled to hydrogenotrophic methanogenesis	−316.3	−235.41
3	$4HPO_3^{2-} + SO_4^{2-} + H^+ \rightarrow 4HPO_4^{2-} + HS^-$	Lithosyntrophic DPO coupled to sulfate reduction	−351.49	−239.27
4	$4H_{2(aq)} + HCO_3^- + H^+ \rightarrow CH_{4(aq)} + 3H_2O$	Hydrogenotrophic methanogenesis	−229.14	−23.77
5	$4H_{2(aq)} + SO_4^{2-} + H^+ \rightarrow HS^- + 4H_2O$	Hydrogenotrophic sulfate reduction	−264.33	−27.63
6	$HPO_3^{2-} + AMP + H^+ \rightarrow H_2 + ADP$	Lithosyntrophic DPO, low energy electrons (ptdF + Hnd + Rnf)	−29.78	−28.13
7	$2HPO_3^{2-} + 2NAD^+ + 2AMP \rightarrow 2NADH + 2ADP$	ptdF	−23.78	−18.57
8	$2ADP \rightarrow ATP + AMP$	Adenylate kinase	+3.70	+3.66
9	$NADH + 2Fd_{red} + 3H^+ \rightarrow NAD^+ + 2Fd_{ox} + 2H_{2(aq)}$	HndABCD	−104.24	−74.26
10	$NADH + 2Fd_{ox}NAD^+ + 2Fd_{red} + H^+$	Rnf	+129.36	+132.64
11	$P_2O_7^{4-} + H_2O \rightarrow 2HPO_4^{2-}$	Pyrophosphatase	−21.66	+15.46
12	$NADH + H^+ \rightarrow NAD^+ + H_{2(aq)}$	NADH oxidation coupled to proton reduction	−6.00	−9.55

Chemical concentration data that were used to calculate activities can be found in [SI Appendix, Table S3](#). Calculated activities can be found in [SI Appendix, Table S4](#).

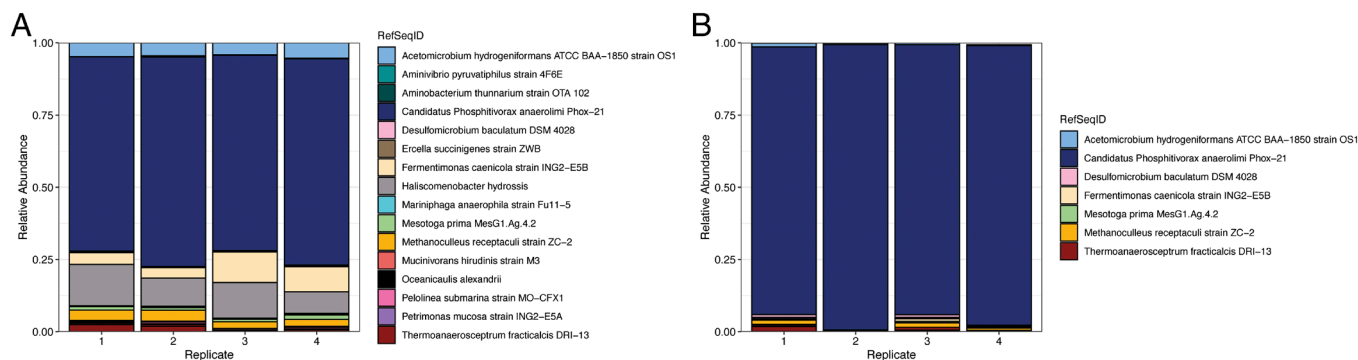


Fig. 1. Relative abundances of microbial taxa from phosphite-oxidizing enrichment cultures (A) with cysteine and (B) with acetate and sulfide. Each bar represents a replicate culture. Taxonomic identification is based on the V4/V5 16S rRNA gene region.

fermentative bacteria (Fig. 1A). Removing CO₂ or cysteine from the medium arrested growth and phosphite oxidation (17, 20). Sulfide alone could not replace cysteine as a reducing agent, which suggested that cysteine, or a degradation product of cysteine, was necessary for phosphite oxidation to occur. Based on the physiology of their closest isolated relatives, we hypothesized that *Acetomicrobium*, *Aminobacterium*, and *Aminivibrio* were likely fermenting cysteine to produce sulfide, ammonia, and pyruvate (21–24). Because organic acids like pyruvate are fermentatively catabolized to acetate, we chose to supplement the media with acetate, a non-fermentable carbon source, rather than pyruvate, to simplify the organic input into the system. By replacing cysteine with acetate (3 mM) and sulfide (1 mM), the community composition was reduced to seven members with the following GTDB taxonomy: Phox-21, *Methanoculleus* sp018433835, *Acetomicrobium hydrogeniformans*, *Desulfomicrobium apsheronum*, *Proteiniphilum massiliensis*, *Mesotoga prima*, and an unclassified member of the class Peptococcia (order DRI-13, genus SV3-B141), with Phox-21 comprising >98% of the community (Fig. 1B). Attempts to isolate Phox-21 on acetoin or crotonate instead enriched for the unclassified strain of Peptococcia (SI Appendix, Fig. S1).

In the sulfide-reduced cultures with acetate, growth reached maximum optical density with concurrent complete oxidation of 10 mM phosphite to 10 mM phosphate after 4 d of incubation

(Fig. 2). Phosphite oxidation was stoichiometric with methane production (~4 mM phosphite oxidized for every 1 mM methane produced) in the methanogenic phosphite-oxidizing culture (Table 1, Rxns. 2 and 4; Fig. 3A). 86 to 98% of electrons could be accounted for in methane, with the remainder likely reflecting analytical uncertainty and anabolic electron sinks. In the presence of the methanogenic inhibitor 2-bromoethanesulfonate (BES, 1 mM), growth and phosphite oxidation of the methanogenic phosphite-oxidizing culture were inhibited (Fig. 2).

Acetate Is Required But Is Not the Source of Methane. Growth and phosphite oxidation required acetate, but acetate consumption was not stoichiometric with phosphite oxidation throughout the growth curve, indicating that acetate serves primarily as a carbon source for biomass rather than an electron donor (SI Appendix, Figs. S2 and S3). Because of the long-term maintenance of a methanogen, *Methanoculleus* sp., in the enrichment, we considered the possibility that acetate was required for acetoclastic methanogenesis. Measurements of methane via GC-MS in cultures incubated with ¹³C bicarbonate or ¹³C acetate showed that carbon from acetate was not the source of methane (SI Appendix, Fig. S4). Methane measured in the condition incubated with ¹³C bicarbonate had a significantly higher enrichment of ¹³C/¹²C compared to the unlabeled control ($P = 2.31 \times 10^{-13}$), while the

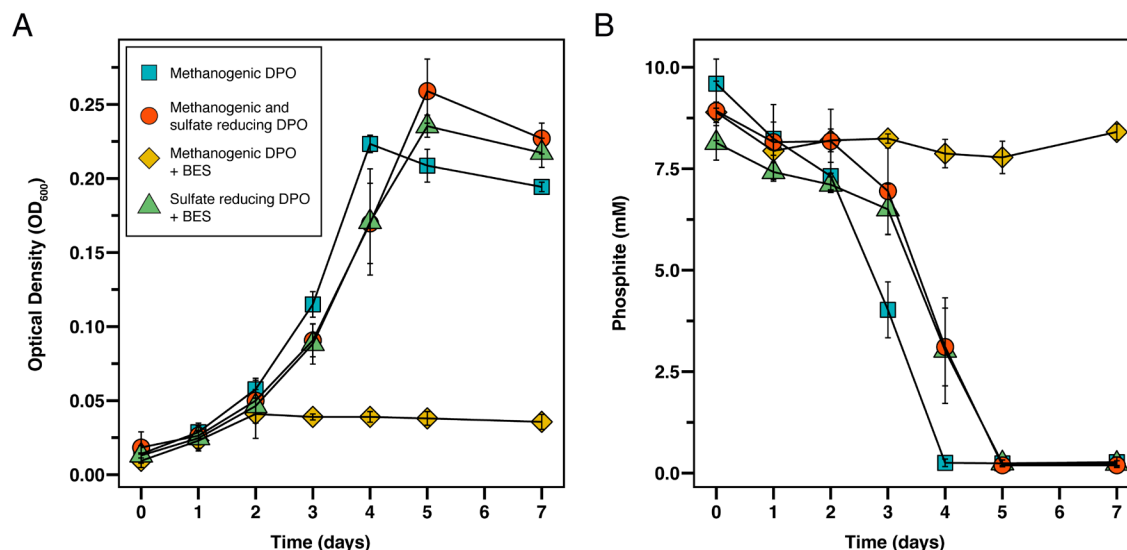


Fig. 2. Growth (A) and phosphite concentration (B) of phosphite-oxidizing enrichment cultures under methanogenic conditions (blue squares), with 1 mM BES (yellow diamonds), methanogenic and sulfate-reducing conditions with *D. desulfuricans* G11 (red circles), and sulfate-reducing conditions with *D. desulfuricans* G11 and 1 mM BES (green triangles). Error bars represent SD of triplicate cultures.

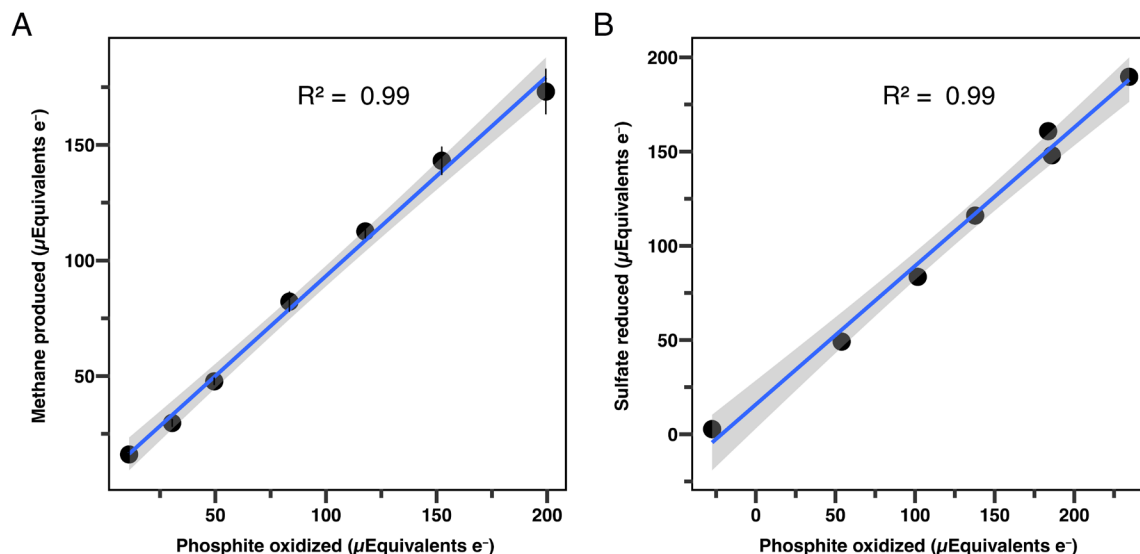


Fig. 3. Relationships between phosphite oxidation and terminal electron-accepting processes under methanogenic conditions (A) or sulfate-reducing conditions (B). Concentrations of phosphite, methane, and sulfate were normalized by μM electron equivalents.

conditions incubated with ^{13}C acetate produced methane that was not significantly enriched compared to the unlabeled control ($P = 0.195$ and 0.386). This was supported by physiological analysis of the *Methanoculleus* sp. isolate, which could only produce methane from H_2/CO_2 or formate, but not acetate (SI Appendix, Fig. S5).

To better understand Phox-21's carbon metabolism and its requirement for acetate, incubations were amended with ^{13}C -labeled bicarbonate or ^{13}C -labeled acetate. Based on analysis of the mass spectra from abundant peptides detected by metaproteomics, carbon from bicarbonate and acetate was incorporated into peptides produced by Phox-21. Within the peptide YDLVEGLK ($z = 2^+$), which matched to the protein sequence of the NAD^+/AMP -dependent phosphite dehydrogenase PtdF/ApdA, there was one carbon substitution from ^{13}C bicarbonate and 15 carbon substitutions from ^{13}C acetate, (SI Appendix, Fig. S6), indicating that Phox-21 assimilates carbon from both bicarbonate and acetate, with acetate serving as the predominant carbon source. This is consistent with the physiological requirement for acetate (SI Appendix, Figs. S2–S4) and the expression of proteins involved in carboxylation reactions (i.e., pyruvate:ferredoxin oxidoreductase, PEP synthase, and PEP carboxylase) detected in the metaproteome, suggesting that the pathway may function in partial CO_2 incorporation (e.g., anaplerotic reactions) rather than supporting full autotrophic growth under conditions tested. The in-depth carbon metabolism of Phox-21 and the relative contributions of CO_2 assimilation vs. acetate assimilation is the subject of current investigations and will be described in a separate publication.

DPO Requires and Maintains Low H_2 Concentrations. If Phox-21 is a syntroph, its metabolism likely depends on interspecies electron transfer to the partner organism, *Methanoculleus* sp. Because *Methanoculleus* sp. produces methane only from H_2/CO_2 and formate (SI Appendix, Figs. S4 and S5), we surmised that Phox-21 couples phosphite oxidation to proton reduction to produce H_2 (Table 1, Rxn. 1). This is consistent with interspecies H_2 transfer, a common form of electron exchange in organosynthetic interactions (3). Hydrogenogenesis from proton reduction is energetically difficult because of the low midpoint redox potential of the H^+/H_2 couple ($E^0 = -414$ mV; equivalent $\Delta G_{37^\circ\text{C}}^0 = +18.6$ kJ/mol) and is thus sensitive to the accumulation of H_2 . The syntrophic partner organism continuously removes

and maintains low H_2 concentrations that are thermodynamically necessary for the reaction to proceed. This mechanism is widely observed in organosynthetic interactions involving methanogens and provided a plausible framework for understanding Phox-21's catabolism and its relationship with *Methanoculleus* sp (Table 1, Rxn. 2).

To confirm that Phox-21 requires a H_2 -consuming partner, we inoculated the culture with the model hydrogenotrophic sulfate reducer *Desulfovibrio desulfuricans* G11 in the presence 1 mM BES (Fig. 2B). Growth and phosphite oxidation were restored, and stoichiometric phosphite oxidation and sulfate consumption (~ 4 mM phosphite oxidized for every 1 mM sulfate reduced) was observed under sulfate-reducing conditions, which was consistent with the syntrophic coupling of phosphite oxidation to hydrogenotrophic sulfate reduction (Table 1, Rxns. 3 and 5; Fig. 3B). 80 to 91% of electrons could be accounted for through sulfate reduction, with the remainder likely reflecting analytical uncertainty and anabolic electron sinks.

Threshold H_2 activities for thermodynamic favorability were calculated from the Gibbs energy of phosphite oxidation coupled to proton reduction (Rxn. 1), hydrogenotrophic methanogenesis (Rxn. 4), hydrogenotrophic sulfate reduction (Rxn. 5), and phosphite oxidation coupled to proton reduction, NAD^+ reduction, and ADP formation (Rxn. 6) as a function of $\text{H}_{2(\text{aq})}$ activity (Fig. 4). Under in vitro conditions of the enrichment culture, calculations predict that hydrogenotrophic methanogenesis is exergonic at a minimum H_2 activity of -8.95 (≈ 1.12 nM $\text{H}_{2(\text{aq})}$), whereas hydrogenotrophic sulfate reduction has a lower threshold H_2 activity (-9.56 , ≈ 0.275 nM $\text{H}_{2(\text{aq})}$) (Fig. 4). This difference is consistent with the lower K_s for H_2 observed in sulfate reducers relative to methanogens (25).

Thermodynamic calculations demonstrated that phosphite oxidation coupled to proton reduction (Table 1, Rxn. 1) remains exergonic across all realistic H_2 concentrations. However, this represents a chemical construct that fails to account for biological energy conservation to drive anabolism. The reaction becomes endergonic only at an aqueous H_2 activity of 0.93 (~ 8.51 M $\text{H}_{2(\text{aq})}$; Fig. 4), a value that far exceeds concentrations attainable in natural or laboratory systems. Thus, phosphite oxidation is effectively uninhibited by H_2 accumulation and should remain exergonic under all environmentally relevant conditions. This reaction yields

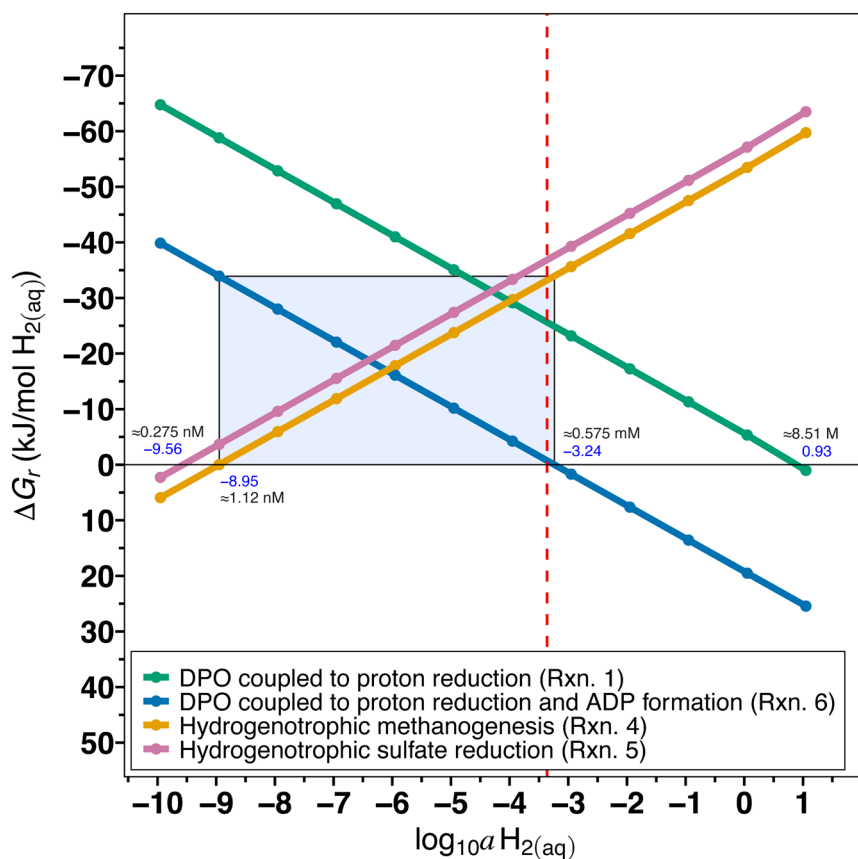


Fig. 4. Gibbs energies, ΔG_r , of reactions 1, 4, 5, and 6 (Table 1) as a function of hydrogen activity. Activities (Dataset S4) for products and reactants were calculated from the concentrations listed in Dataset S3. The blue box indicates the range of hydrogen activities where hydrogenotrophic methanogenesis and DPO coupled to proton reduction and ADP formation are both exergonic. The red dashed line indicates the aqueous hydrogen activity equivalent to adding 10 mL of hydrogen gas to the headspace of a culture tube. ΔG_r was calculated using activities determined from the following concentrations: 10 mM phosphite, 100 μ M methane, 10 mM sulfate, 2.2 mM ATP, 0.83 mM ADP, 0.3 mM AMP, and 100 μ M NAD⁺ and NADH at 37 °C.

–52.9 kJ/mol H₂ under the in vitro H₂ concentrations measured in the culture (8.2 ± 0.7 nM H_{2(aq)}; SI Appendix, Fig. S7). Although the net reaction of phosphite oxidation coupled to proton reduction is exergonic at the measured H₂ concentrations, growth and phosphite oxidation were unexpectedly inhibited when excess H₂ was added to the headspace of the tubes (~ 0.433 mM H_{2(aq)}; Fig. 5). This inhibition likely reflects a thermodynamic inhibition of intracellular electron-transfer reactions within the cell. When H₂ accumulates, the reactions that couple NADH and reduced ferredoxin oxidation at the hydrogenase approach thermodynamic equilibrium and become endergonic. As a result, excess H₂ limits electron flow through these redox steps and inhibits overall phosphite oxidation and growth, despite the exergonic net reaction.

A Model of Lithosynthetic Phosphite Oxidation in Phox-21. To investigate the metabolic basis for this thermodynamic inhibition, we constructed a model of Phox-21's catabolism using the most abundant peptides detected by metaproteomics (Fig. 6 and Dataset S5). The most abundant protein was the NAD⁺/AMP-dependent phosphite dehydrogenase, PtdF/ApdA, which is part of the ptx-ptd gene cluster conserved in all DPOM (17, 26, 27). Other abundant proteins included the putative phosphite-phosphate antiporter PtdC; the NADH- and ferredoxin-oxidizing, electron confurcating/bifurcating hydrogenase HndABCD; a membrane-bound, Na⁺-translocating ferredoxin:NAD⁺ oxidoreductase Rnf complex, an adenylate kinase; a K⁺-stimulated pyrophosphate-energized Na⁺/H⁺ pump; and ATP synthases. Note that HndC was annotated as a homolog of NADH:quinone oxidoreductase (NuoF) (28). Together, the Rnf complex, electron-confurcating hydrogenase, and ATP generation through substrate-level phosphorylation represent canonical energy conservation mechanisms associated with syntrophic metabolism and form the basis of our proposed model of phosphite oxidation in Phox-21 (Fig. 6) (1, 29, 30).

Proteins involved in carbon assimilation (pyruvate:ferredoxin oxidoreductase, PEP synthase, and PEP carboxykinase) were also abundant, consistent with carbon assimilation from acetate and oxaloacetate.

In our proposed model of lithosynthetic phosphite oxidation (Fig. 6), phosphite first enters the cell via PtdC. Electrons from phosphite are transferred by PtdF/ApdA to NAD⁺ to generate NADH and phosphate, with concomitant AMP phosphorylation to ADP (Table 1, Rxn. 7) (27). This reaction, which accounts for the energy required to form a high-energy phosphoanhydride bond in ADP, yields –18.6 kJ/mol. The resulting ADP pool is converted to ATP + AMP by adenylate kinase (Table 1, Rxn. 8). Reduced ferredoxin required for H₂ generation is produced by the Rnf complex (Table 1, Rxn. 10) at the expense of NADH and a

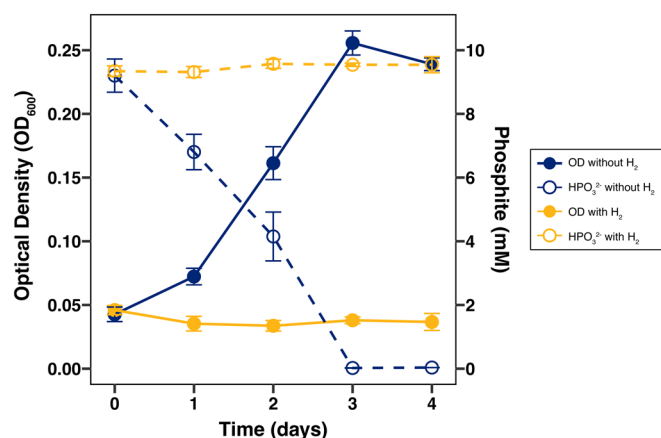


Fig. 5. Growth (solid points and lines) and phosphite concentrations (open points, dashed lines) in the presence and absence of 10 mL H₂ added to the headspace of the culture tubes. Error bars represent SD of triplicate cultures.

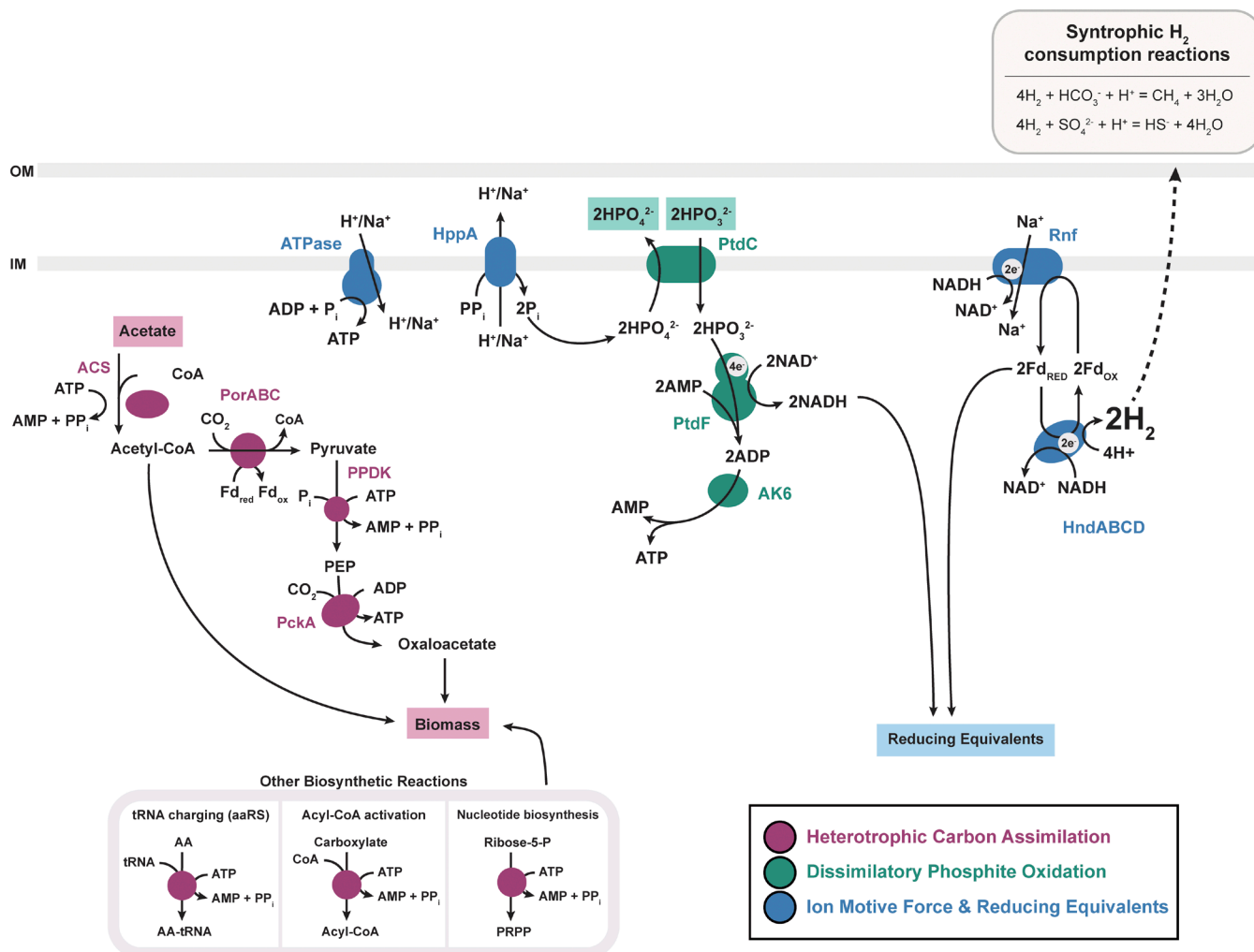


Fig. 6. Metabolic model of dissimilatory phosphite oxidation coupled to proton reduction in *Ca. Phosphitivorax anaerolimi* Phox-21. Heterotrophic carbon assimilation (magenta) includes Acs (acetyl-coenzyme A synthetase), PorABC (pyruvate synthase), and PPK (pyruvate phosphate dikinase), PckA (phosphoenolpyruvate carboxykinase). Dissimilatory phosphite oxidation (green) proceeds via PtdC (phosphite/phosphate antiporter) and PtdF (AMP-dependent phosphite dehydrogenase), which couples phosphite oxidation to phosphorylation of AMP, generating ADP and reducing equivalents. AK6 (adenylate kinase) converts two ADP into ATP and AMP; because PtdF generates one ADP per phosphite oxidized, additional AMP must be supplied from the cytosolic phosphate energy carrier pool. Ion motive force and reducing equivalent generation (blue) involves HppA (K^+ -stimulated pyrophosphate-energized Na^+ pump), Rnf (membrane-bound Na^+ -translocating ferredoxin: NAD^+ oxidoreductase), and HndABCD (NADH- and ferredoxin-oxidizing, electron-confurcating hydrogenase). Pyrophosphate (PP_i) is generated predominantly during biosynthetic ATP/AMP reactions, including tRNA charging, carboxylate activation, and nucleotide biosynthesis, and is subsequently hydrolyzed by HppA to drive ion translocation and support ATP synthesis. The bold dashed arrow indicates diffusive export of H_2 to the syntrophic partner, representing interspecies H_2 transfer.

Na^+ motive force (31, 32). HndABCD, a cytoplasmic FeFe electron-confurcating hydrogenase, then couples the oxidation of reduced ferredoxin and NADH to proton reduction to generate H_2 (Table 1, Rxn. 9) (28, 33, 34). Pyrophosphate generated primarily during ATP-dependent ligation/activation reactions, including tRNA charging, carboxylate activation, and nucleotide biosynthesis, is hydrolyzed by the ion-translocating pyrophosphatase into two molecules of inorganic phosphate (Table 1, Rxn. 11), coupling pyrophosphate hydrolysis to ion translocation and maintenance of the electrochemical gradient. Inorganic phosphate is transported out of the cell by PtdC.

We examined the energetics of individual reactions in each step of our proposed model to understand the thermodynamic inhibition observed in the presence of excess H_2 . The energy yield of reducing protons with NADH to form H_2 (Table 1, Rxn. 12) is close to thermodynamic equilibrium and is thus very sensitive to the accumulation of H_2 ($\Delta G_{12} = -9.6$ kJ/mol H_2 at 10 nM $H_{2(aq)}$). This thermodynamic barrier is overcome in organosyntrophic bacteria by coupling the exergonic oxidation of ferredoxin to the

endergonic oxidation of NADH to generate H_2 in an electron confurcation reaction (Table 1, Rxn. 9) (30, 34). Reduced ferredoxin is generated by the Rnf complex in an energy consuming reaction (Table 1, Rxn. 10; $\Delta G_{10} = +132.6$ kJ/mol) at the expense of NADH and a Na^+ motive force (31, 32). Using electrons donated from ferredoxin and NADH, HndABCD reduces protons to form H_2 . Under Phox-21's intracellular conditions, this reaction (Table 1, Rxn. 9) is exergonic ($\Delta G_9 = -37.1$ kJ/mol H_2 at 10 nM $H_{2(aq)}$).

The key catabolic reaction of this model catalyzed by PtdF/ApdA, HndABCD, and the Rnf complex, which together link phosphite oxidation to H_2 production and energy conservation via substrate-level phosphorylation (Table 1, Rxn. 6; Fig. 6), yields -28.1 kJ/mol H_2 and becomes endergonic at an aqueous H_2 activity -3.24 (≈ 0.575 mM $H_{2(aq)}$; Fig. 4). Thus, while the net reaction of phosphite oxidation coupled to proton reduction is exergonic, the sensitivity of the hydrogenase to H_2 concentrations presents a thermodynamic bottleneck that explains Phox-21's requirement for a H_2 -consuming syntrophic partner. For both phosphite

oxidation and hydrogenotrophic methanogenesis to remain exergonic, H₂ concentrations in the cultures must be maintained at a minimum of 1.12 nM to sustain methanogenesis, and at a maximum of 0.575 mM for phosphite oxidation to proceed (Fig. 4). This determination is consistent with DPO inhibition in our cultures when excess H₂ was added to the headspace of the culture tubes (Figs. 4 and 5).

Discussion

Candidatus Phosphitivorax Anaerolimi Phox-21 Is an Obligate Lithosynthetic Phosphite Oxidizer. Dissimilatory phosphite oxidation in *Candidatus Phosphitivorax anaerolimi* Phox-21 represents an unusual mode of energy metabolism in which the catabolic oxidation of phosphite to phosphate is directly linked to energy conservation within a single enzymatic step. The AMP-dependent phosphite dehydrogenase PtdF couples phosphite oxidation to energy conservation by forming a high-energy phosphoanhydride bond through the direct phosphorylation of AMP to ADP. This mechanism is conceptually analogous to substrate-level phosphorylation in organosynthetic bacteria, where energy is conserved directly during catabolism rather than exclusively via ATP synthase. We propose that Phox-21's energy metabolism and physiological dependence on a hydrogenotrophic methanogen represent a previously undescribed mode of obligate syntrophic interaction, which we have designated as lithosynthesis, where the electron donor supporting interspecies H₂ transfer in a methanogenic syntrophic interaction is inorganic, rather than organic. This expands the conceptual framework of interspecies H₂ transfer-based syntrophy beyond organotrophic interactions and motivates future work to examine whether a broader spectrum of similar interactions occur in anoxic ecosystems.

Several lines of evidence support the conclusion that Phox-21 shares a lithosynthetic relationship with *Methanoculleus* sp. Phox-21's resistance to isolation with traditional isolation methods or with substrates that have been successful for isolating facultative organosynthesizers (SI Appendix, Fig. S1), metabolic specialization, phylogenetic relatedness to fastidious organosynthesizers (SI Appendix, Figs. S8 and S9), indicated that Phox-21 was physiologically distinct from previously isolated DPOM FiPS-3 and DYL19 (17, 18, 35–37). The long-term coenrichment of the methanogen *Methanoculleus* sp. (Fig. 1), the indirect inhibition of growth and phosphite oxidation with the specific methanogenic inhibitor BES (Fig. 2), the direct thermodynamic inhibition of hydrogenogenesis with elevated H₂ concentrations, and the linear correlation of phosphite oxidation and methanogenesis or sulfate reduction (Fig. 3) demonstrated that Phox-21 requires a syntrophic partner that can maintain low H₂ concentrations. Notably, the specific coenrichment of *Methanoculleus* sp. over other hydrogenotrophic methanogens was also suggestive of an obligate syntrophic relationship: members of the genus *Methanoculleus* have a high affinity for and may be adapted to low fluxes of H₂ produced through syntrophic oxidation, and often associate with obligately syntrophic acetate oxidizers and other species of *Phosphitivorax* (38–45).

These observations are consistent with the physiological differences between Phox-21 and FiPS-3 and DYL19, and aspects of our proposed model are corroborated by the metabolic model proposed by Mao and colleagues (27). Notably, the initial use of cysteine as a reducing agent inadvertently enabled the enrichment of a mixotrophic DPOM. We initially assumed that, like FiPS-3 and DYL19, Phox-21 was a lithoautotroph coupling phosphite oxidation to carbon fixation, which was supported by the presence of the reductive glycine pathway for carbon fixation in Phox-21's

genome and necessity of CO₂ in the growth culture (17, 18). However, our current study demonstrates that growth and phosphite oxidation required acetate (SI Appendix, Figs. S2 and S3), which was likely initially derived from cysteine catabolism. Phox-21's dependence on acetate can be explained by both genomic and physiological evidence. While its genome encodes acetyl-coA synthetase (ACS), which enables acetate assimilation into acetyl-coA, it lacks phosphoacetyltransferase (PTA) and acetate kinase (ACK), the enzymes that generate acetate and ATP via substrate-level phosphorylation in the reductive glycine pathway (19). As a result, Phox-21 cannot conserve energy through acetogenesis and instead relies on external acetate for biosynthesis. Proteomic and stable isotope probing indicate that the reductive glycine pathway remains active (SI Appendix, Fig. S6 and Dataset S5), likely serving in small amounts of CO₂ assimilation rather than as the primary route of carbon fixation. While the reductive glycine pathway may contribute to carbon incorporation from CO₂, it cannot account for energy conservation or endogenous acetate generation. Thus, we could not confidently assign CO₂ as the terminal electron acceptor for phosphite in a single catabolic reaction.

We propose that H⁺, not CO₂, is the electron acceptor for phosphite and that acetate is required as carbon source for biosynthesis (Fig. 6). In FiPS-3 and DYL19, NADH generated by ptdF serves as the reducing power for lithoautotrophic CO₂ fixation in the Wood–Ljungdahl pathway. The acetate produced by this pathway supports carbon assimilation. Because the reducing equivalents generated during DPO are consumed by CO₂ and/or sulfate reduction, these organisms may not require a syntrophic partner. In contrast, Phox-21 assimilates carbon primarily from acetate and must dispose of excess reducing equivalents through interspecies H₂ transfer. Thus, Phox-21's dependency on a syntrophic partner may reflect both its inability to produce acetate through an NADH-consuming process and its need to offload H₂ generated during DPO.

Surprisingly, ptxD, the NAD⁺-dependent phosphite dehydrogenase assumed to be involved in both assimilatory phosphite oxidation (APO) and DPO (17, 26, 46, 47), was not detected in the most abundant peptides in our metaproteomic analysis. In concurrence with the results from a previous study on DYL19 and FiPS-3, ptdF/apdA, the NAD⁺/AMP-dependent phosphite dehydrogenase appears to be the primary enzyme involved in DPO (27), and indeed, PtdF was the most abundant peptide detected. In our proteomics results, PtdF and PtdC were the only enzymes encoded by the ptx-ptd gene cluster in the most abundant peptides produced by Phox-21 (Dataset S5). This raises the question of why evolution has conserved genes encoding two NAD⁺-dependent phosphite dehydrogenases in the same gene cluster in all identified DPOM. Perhaps PtxD is mainly used for assimilatory phosphite oxidation and can be expressed by DPOM when phosphate is extremely limiting or it may be required to maintain a NAD⁺/NADH redox balance to support anabolic reactions.

Ecological and Biogeochemical Implications of Lithosynthetic DPO. Organosynthetic interactions facilitated by interspecies electron transfer via diverse mechanisms, including interspecies formate transfer, direct interspecies electron transfer (DIET), and quinone-mediated interspecies electron transfer (QUIET) are key to the methanogenic degradation of complex organic matter (1–3, 48). Syntrophic bacteria often couple the oxidation of fermentation intermediates such as organic acids and alcohols to hydrogenogenesis through proton reduction. Even under optimal growth conditions, syntrophic catabolic reactions yield very little energy, and are so thermodynamically constrained that the

accumulation of small amounts of H_2 can shift the energetics of the reaction from exergonic ($\Delta G_r < 0$) to endergonic ($\Delta G_r > 0$) (2, 49). These thermodynamic constraints are overcome by the consumption of H_2 by a syntrophic partner, which is often a methanogenic archaeon. Interspecies H_2 transfer allows for the complete degradation of fermentation intermediates that neither syntrophic partner can degrade alone.

In lithosyntrophy, phosphite oxidation drives hydrogenotrophic methanogenesis (Fig. 7A). The identification of lithosyntrophic phosphite oxidation reveals a previously unrecognized mechanism by which inorganic compounds contribute to methanogenesis and carbon cycling in anoxic environments. Microbial methane production is canonically assumed to be driven by complex organic matter degradation to H_2/CO_2 , formate, acetate, and methylated compounds (1, 48, 50) or by geologically produced H_2 in subsurface environments (51–53). However, the finding of lithosyntrophic phosphite oxidation demonstrates that inorganic compounds can serve as an energy source in obligate syntrophic interactions, through the coupling of phosphite oxidation to hydrogenotrophic methanogenesis (Table 1, Rxns. 1 and 2). This expands the known metabolic diversity of syntrophic interactions and establishes an unrecognized link between the phosphorus and carbon cycles. Given that archaeal methanogenesis in anoxic environments is estimated to produce more than half of annual methane emissions (54), lithosyntrophy may serve as a major but previously unaccounted driver of methanogenesis. These processes warrant further investigation across diverse environments.

A key outstanding question is where and how phosphite is produced in present day environments, which limits our understanding of the environmental distribution and ecological relevance of lithosyntrophic DPOM. FiPS-3, DYL19, and Phox-21 were isolated or enriched from engineered environments (wastewater treatment sludge) or impacted environments (Canale Grande, Venice, Italy). In wastewater treatment sludge, phosphite concentrations have been measured at 0.157 to 0.278 μM (55, 56). Few measurements of phosphite in natural environments have been reported (57, 58), but in freshwater samples from Florida (rivers, ponds, and swamps),

reduced oxidation state phosphorus (phosphite and hypophosphite) accounted for 10 to 20% of the total phosphorus pool (59). Higher phosphite concentrations were detected in the stagnant waters (ponds/swamps) compared to the river samples. The pond environments in this study are impacted by anthropogenic activity, while the swamp represents a natural environment. Like wastewater treatment sludge, the stagnant waters where phosphite was detected by Pasek et al. are likely anoxic and nutrient rich. The source of phosphite in these systems is unclear, but it may be produced through phosphate reduction or phosphonate degradation in the course of organic matter catabolism in high carbon, reducing environments (Fig. 7B).

In marine and terrestrial serpentinite muds, phosphite was detected at 23 to 60% of total P (50 to 100 μM) (60). The process of serpentinization generates H_2 which can react with CO_2 to produce fluids enriched in small organics like acetate (61–63). ptdF was detected in metagenomic datasets from serpentinizing sites at the Lost City Hydrothermal Field, suggesting that microorganisms with the potential for DPO exist at these sites (64). While the presence of acetate, phosphite, and the key marker gene for DPO are promising, serpentinizing fluids are characterized by high H_2 concentrations, with concentrations measured at up to 15 mM at the Lost City Hydrothermal Field (65, 66). While it is possible that acetogenic DPO could serve as a microbial metabolism at serpentinizing sites, high background H_2 concentrations in vent fluids would likely be inhibitory to lithosyntrophic DPOM (Fig. 7B).

In nutrient-limited environments, such as deep subsurface sediments, lithosyntrophic phosphite oxidizers may play important ecological roles in transferring both reducing equivalents (in the form of H_2) and bioavailable phosphate to their surrounding microbial communities. Because phosphate has been a limiting nutrient for primary production across geologic time (67, 68), the ability to perform DPO, particularly in environments where phosphite is the dominant phosphorus species, may give lithosyntrophic DPOM a competitive advantage, potentially acting as keystone species that shape microbial ecosystem structure and function.

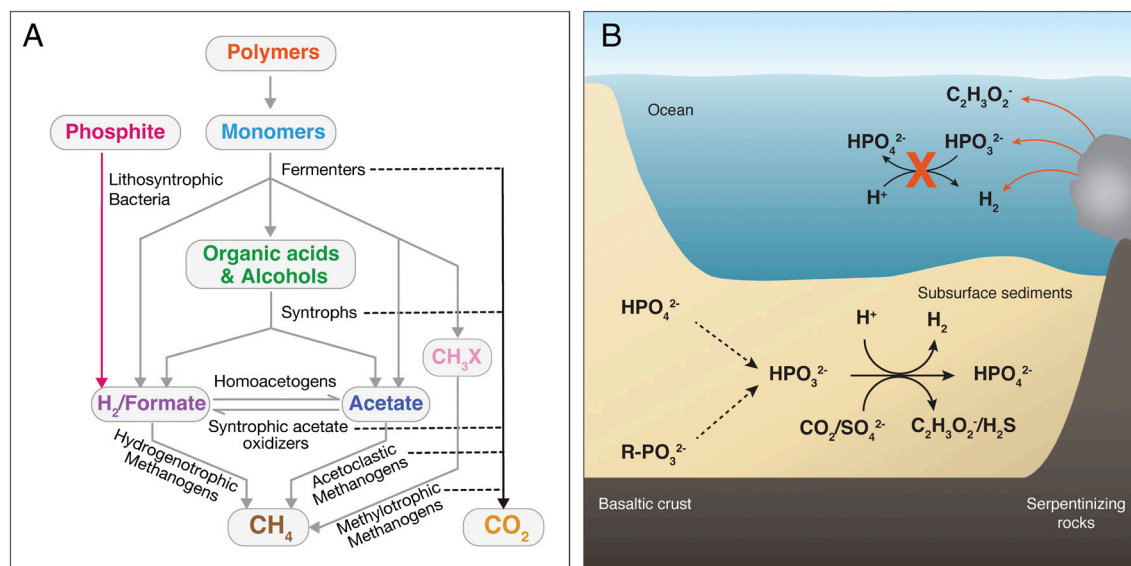


Fig. 7. Geochemical and ecological contexts of lithosyntrophic phosphite oxidation. (A) Electron flow in organotrophic and lithosyntrophic carbon degradation. While organosyntrophy drives methanogenesis through hydrogen production from organic acids or alcohols, lithosyntrophy similarly supports methanogenesis via hydrogen production from phosphite oxidation. (B) Mechanisms of phosphite production and geochemical conditions that may support or inhibit lithosyntrophic DPO. Serpentinization produces hydrogen, phosphite, acetate, and highly reducing conditions, but high hydrogen concentrations are inhibitory to lithosyntrophic DPO. In anoxic marine sediments, phosphite may be produced through phosphate reduction or phosphonate degradation and can be oxidized through lithosyntrophic DPO, acetogenic DPO, or sulfidogenic DPO. Red lines indicate abiotic processes, dotted lines indicate hypothetical processes, and solid black lines indicate known microbial metabolic reactions.

Phosphite has been detected in early Archean sedimentary carbonates (69), early Archean iron-rich metasedimentary rocks (70), and late Archean banded iron formations (71), and it has been proposed that under highly reducing conditions prior to the widespread oxygenation of Earth's atmosphere, phosphite represented a more abundant and bioavailable phosphorus source. Phylogenomic reconstructions of the ptx-ptd gene cluster estimate that DPO emerged ~3.2 Ga in the most recent common ancestor of the Firmicutes and Desulfobacterota (17). While molecular clock analyses by Boden et al. estimate the earliest evolution of DPO in the Firmicutes around 2.3 Ga, these predictions were based solely on the presence of ptxD, which does not appear to be involved in DPO (72). Instead, ptdF and ptdC, which were detected in high abundance in the proteomes of Phox-21, FiPS-3, and DYLL19, should be considered as the key marker genes of DPO, while ptxD is likely primarily indicative of assimilatory phosphite oxidation (27). Further investigations on phosphorus redox cycling and on the evolution of microbial phosphite oxidation are required to constrain the role of this metabolism in Earth's history. Integrating geochemical, ecological, and evolutionary perspectives will be key to resolving its broader geobiological significance.

Conclusion

These findings broaden our understanding of the role of inorganic compounds in anaerobic carbon cycling by revealing previously unrecognized lithosynthetic interactions and point to the potential existence of similar interactions involving other inorganic substrates. The demonstration that phosphite oxidation can drive hydrogenotrophic methanogenesis redefines the role of phosphorus in anoxic ecosystems and establishes a metabolic and biogeochemical link between the phosphorus and carbon cycles. Lithosynthetic DPO may act as keystone species in certain environments, serving as producers of reducing equivalents and sources of bioavailable phosphorus. Together, these results suggest that lithosynthetic DPO may play a critical role in supporting primary production, driving methanogenesis, and redistributing bioavailable phosphorus in anoxic, high-carbon environments, engineered systems, electron acceptor-limited environments, the modern deep biosphere, and potentially in early Earth environments.

Methods

Thermodynamic Calculations. The Gibbs energy of reaction ΔG_r was calculated for 12 reactions involved in the DPO enrichment culture (Table 1 and Dataset S1), using

$$\Delta G_r = \Delta G_r^0 + RT \ln Q_r, \quad [1]$$

where ΔG_r^0 is the standard state Gibbs energy, R refers to the universal gas constant, T represents the temperature in Kelvin, and Q_r denotes the reaction quotient. Values of ΔG_r^0 were calculated for each reaction at 37 °C and 1 bar using the revised Helgeson-Kirkham-Flowers (HKF) equations of state (73–75) using the "subct" command from the R software package CHNOSZ v2.1.0 (76). Thermodynamic data in CHNOSZ are sourced from the OrganoBioGeoTherm database, which are derived from a number of sources documented in <https://chnosz.net/download/refs.html>. Values represented by the activity quotient, Q_r , were calculated with

$$Q_r = \prod a_i^{v_{i,r}}, \quad [2]$$

where a_i corresponds to the activity of the i th species raised to its stoichiometric coefficient, $v_{i,r}$, in the r th reaction, which is positive for products and negative for reactants. Activities were calculated from the chemical composition of the cultivation medium (Dataset S2, described below) with a range of $H_{2(aq)}$ concentrations using the "ae.speciate" command from the aqueous speciation package AqEquil v0.18.0 (Datasets S3 and S4) (77). Intracellular concentrations of ATP (2.2 mM),

ADP (0.83 mM), and AMP (0.3 mM) were sourced from ref. 78. Intracellular concentrations of reduced and oxidized ferredoxin, NAD^+ , and NADH were assumed to be 100 μ M. The standard state Gibbs energy for ferredoxin oxidation was calculated from its redox potential (79). Henry's Law constants ($\text{mol L}^{-1} \text{atm}^{-1}$) for hydrogen and methane were calculated at 37 °C using the equilibrium constants of the gas dissolution reaction for each compound:

$$CH_{4(g)} \leftrightarrow CH_{4(aq)}, K_H = 1.01 \times 10^{-3} \text{ mol L}^{-1} \text{atm}^{-1},$$

$$H_{2(g)} \leftrightarrow H_{2(aq)}, K_H = 7.47 \times 10^{-4} \text{ mol L}^{-1} \text{atm}^{-1}.$$

Media and Cultivation Conditions. Highly enriched phosphite-oxidizing enrichment cultures were originally inoculated and maintained by ref. 17. Briefly, anoxic, HEPES buffered basal medium (Dataset S2) was prepared and dispensed under N_2/CO_2 (80:20). All chemicals were sourced from Sigma-Aldrich. Medium amendments were added to the basal medium from sterile, anoxic 1 M stocks. The medium was reduced with 3 mM L-cysteine HCl and 10 mM sodium phosphite dibasic pentahydrate was added as the electron donor and sole phosphorus source. Enrichment cultures were spiked with an additional 10 mM sodium phosphite after 2 wk of growth at 37 °C and passaged with a 10% inoculation volume after 1 mo. Growth was measured as optical density at 600 nm (OD_{600}) with a Genesys 20 Visible spectrophotometer (Thermo Scientific).

L-cysteine HCl was replaced with 3 mM sodium acetate and 1 mM sodium sulfide nonahydrate. For isolation experiments, phosphite was replaced with 10 mM sodium phosphate monobasic and the medium was amended with 10 mM crotonic acid or acetoin from sterile, anoxic stock solutions. For conditions with excess H_2 , 10 mL of H_2 gas was added to the headspace of a Balch tube containing 10 mL of culture. Methanogenesis was inhibited with 1 mM 2-bromoethanesulfonate (BES). The methanogenic DPO culture was grown under sulfate-reducing conditions (10 mM sulfate) in the presence of BES by adding the model hydrogenotrophic sulfate reducer, *Desulfovibrio desulfuricans* G11 at 50% of the inoculum.

Isolation and Characterization of Methanoculleus sp. To isolate the methanogen, *Methanoculleus* sp., the anoxic basal medium was amended with 10 mM sodium phosphate monobasic and H_2 was added at 20% of the headspace volume. Methane was measured as described below and the culture was isolated by serial dilution to extinction. Sample purity was assessed by gene rRNA amplicon sequencing and metagenomic sequencing, as described below. To determine which methanogenic substrates could be used, *Methanoculleus* was grown at 37 °C with 20:80 H_2/CO_2 (shaking at a slant, 250 rpm), with 10 mM sodium formate, or with 10 mM sodium acetate.

Sampling and Measurements of Ions and Gases. 300 μ L culture was collected and filtered through a 0.2 μ m nylon syringe filter for analytical measurements and stored at 4 °C until analysis. Phosphite, phosphate, and sulfate concentrations were measured via ion chromatography using a Dionex ICS 2100 with a Dionex IonPac AS 16 column (4 $\times$ 250 mm) (Thermo Fisher Scientific) as described in ref. 18. Headspace methane concentrations were measured via gas chromatography using an Agilent gas chromatograph model 7890A (G3440A) with a Supelco SP-2380 Fused Silica Capillary column. Methane was measured using a flame ionization detector (FID) at 300 °C, a column temperature of 50 °C. Headspace hydrogen concentrations were measured by gas chromatography with a reducing compound photometer using a Peak Performer 1 reductive gas analyzer (Peak Laboratories) with a bed temperature of 265 °C, a column temperature of 105 °C, and N_2 as the carrier gas.

Stable Isotope Probing and Metaproteomics. 50 mL cultures were amended with either 10 mM ^{13}C -sodium bicarbonate (Sigma Aldrich), 3 mM sodium acetate-2- ^{13}C , or 3 mM sodium acetate-2- ^{13}C without CO_2 and bicarbonate, and incubated for 1 wk at 37 °C. ^{13}C incorporation into headspace methane and CO_2 was measured by gas chromatography-mass spectrometry (GC-MS; Agilent) with a GasPro column on day 0 and day 7 of incubation. The ratio of $^{13}CH_4$ to $^{12}CH_4$ on day 7 was calculated and significance testing was performed using ANOVA and Tukey's honestly significant difference (HSD) test.

After 1 wk of incubation, the entire culture (50 mL) was pelleted for tryptic digestion by centrifuging at 16,873 relative centrifugal force (rcf) for 10 min. The pellet was washed and resuspended in 100 mM NH_4HCO_3 , pH 7.5, and transferred to a microcentrifuge tube. The pellet was centrifuged again and resuspended in 50 μ L

of 100 mM NH_4HCO_3 . 25 μL of 0.2% RapiGest SF surfactant (Waters 186001861) was added to make the proteins more accessible for alkylation and digestion and the samples were heated for 15 min at 80 °C. 2.5 μL of 100 mM dithiothreitol (15.4 $\mu\text{g}/\mu\text{L}$) was added and the samples were heated for 30 min at 60 °C. The samples were cooled to room temperature and 7.5 μL of 100 mM iodoacetamide (55.5 $\mu\text{g}/\mu\text{L}$) was added to alkylate cysteines. The tubes were incubated at room temperature in the dark for 30 min. 5 μL of Promega trypsin (1 $\mu\text{g}/\mu\text{L}$) was added and the samples were incubated overnight at 37 °C. Following the trypsin digestion, 10 μL of 5% trifluoroacetic acid was added to hydrolyze the RapiGest. The samples were incubated for 90 min at 37 °C, then centrifuged for 30 min at 14,000 RPM at 4 °C. 30 μL of the supernatant was transferred to a sample vial.

Triplicate trypsin-digested protein samples of unlabeled controls, ^{13}C -bicarbonate, and ^{13}C -acetate were analyzed with an Acquity M-class ultraperformance liquid chromatography (UPLC) system that was connected in line with the Synapt G2-Si mass spectrometer (Waters, Milford, MA) as described previously (80). Data were acquired and analyzed using MassLynx (v4.1, Waters) and Progenesis Q1 for Proteomics software (v4.2, Waters Nonlinear Dynamics). Data were searched against the translated protein sequence of the metagenome (described below).

DNA Extraction, Amplicon and Metagenomic Sequencing, and Bioinformatics. 1.5 mL of culture was collected for DNA extraction and centrifuged for 10 min at 16,873 rcf. Supernatant was removed and pellets were stored at -20 °C until DNA extraction. DNA was extracted for 16S rRNA gene community sequencing using enzymatic digestion followed by ethanol precipitation. Pellets were resuspended in 200 μL enzymatic lysis buffer (20 mM Tris-HCl pH 8.0, 2 mM sodium EDTA, 1.2% Triton X-100) and transferred to a 96-well plate. 20 μL Proteinase K (20 mg/mL) and 200 μL of 4 M guanidine HCl was added to each well and plates were incubated overnight at 55 °C. 4 μL of RNase A (100 mg/mL) was added to each well and plates were incubated for 2 h at room temperature. 200 μL of 100% ethanol was added and the contents of each well were transferred to a 96-well plate filter. Columns were washed with 400 μL of 70% ethanol and DNA was eluted with 100 μL of TE buffer (100 mM Tris, 10 mM EDTA, pH 8.0). The V4/V5 16S rRNA gene region was amplified via PCR using the 515F/926R primers (81), but with in-line dual Illumina indexes (82, 83). Amplicons were sequenced on an Illumina MiSeq (Illumina, San Diego, CA, USA) with 2 $\times$ 300 bp Illumina v3 reagents. Custom Perl scripts implementing PEAR (84) were used for read merging, USearch (85) for filtering reads with more than one expected error and demultiplexing using inline indexes, and UNoise (86) for filtering rare reads and chimeras. 16S rRNA gene sequences in the relative abundance table were searched against the NCBI RefSeq database and matched metagenome-assembled genomes (MAGs, described below) to assign taxonomy.

DNA was extracted for metagenomic sequencing using the Qiagen DNeasy UltraClean DNA extraction kit following the manufacturer's protocols. Metagenomic libraries were prepared and sequenced on a NovaSeqX platform by Novogene

Corporation Inc. Metagenomic raw reads were trimmed using Sickle (87) to remove low quality reads ($q < 20$). Triplicate metagenomes (biological replicates) were coassembled using MEGAHIT (88) and contigs shorter than 1,000 base pairs were eliminated using the "anvi-script-reformat-fasta" command from the software package anvi'o (89). Raw reads were mapped to the assembly using bowtie2 and sorted and indexed using samtools (90, 91). Genome binning of assembled reads was performed using MetaBat, MetaBAT 2 (92), concoct (93), and MaxBin 2.0 (94). An optimized, nonredundant set of MAGs was selected using DAS Tool (95). MAG quality was assessed using CheckM2 (96) and taxonomic identity was determined using GTDBTK v2.4.1, release 226 (97). MAG genome sequences were translated to protein sequences using Prokka (v1.14.6) (98), and protein sequences for each MAG were concatenated into a single file to use for peptide identification from the metaproteome.

A phylogenomic tree of the phylum Desulfobacterota was constructed using Phox-21 MAGs from this study and from (17, 18) using the GTDBTK de novo workflow (97). A 16S rRNA gene sequence phylogenetic tree of Phox-21, *Desulfotignum phosphitoxidans* FIPS-3, *Phosphitispora fastidiosa* DYLL19, and Phox-21's closest relatives was constructed using the top 1,000 sequence matches from the NCBI core nucleotide (core_nt) database and the top 10 sequence matches from the NCBI 16S rRNA gene type strain database. The 16S rRNA gene sequence for *Sulfurovum aggregans* was used as an outgroup. Sequences were dereplicated using the rmdup command from SeqKit (99), aligned using MUSCLE (100), trimmed using trimAl (101), and a tree was constructed using IQ-Tree (102). Trees were visualized using iTOL (103).

Data, Materials, and Software Availability. Metagenomic reads for the enrichment culture containing cysteine, metagenomic reads and 16S rRNA gene short read sequences for the enrichment culture containing acetate and sulfide data have been deposited in NCBI under accession numbers PRJNA655520 (104) and PRJNA1305015 (105). All other data are included in the article and/or supporting information.

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